

Digital Banking Quality Service and Customers' Perception (Lebanese Banks)

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Abstract

This research investigated how digital banking service dimensions—tangibility, reliability, responsiveness, assurance, and empathy—affect customers of Lebanese banks. Data were collected using descriptive statistical methods through a five-section questionnaire administered to a random sample of bank customers across Lebanon. The collected responses were analyzed and hypotheses tested with the SPSS software. Findings revealed a strong overall correlation among the five dimensions of digital banking quality service and customers' perception and engagement. Moreover, tangibility, responsiveness, assurance, and empathy showed significant influence on customer engagement, whereas reliability did not demonstrate a notable effect. There was no effect of the demographic variable (the number of years of dealing with the bank) on satisfaction. The study came up with suggestions and recommendations, including marketing more for the mobile application, bills, and other electronic services provided by the banks and supporting the appropriate use of e-banking to encourage customers to use them more.

Keywords: digital banking, tangibility, responsiveness, assurance, reliability, and empathy

1. Introduction

1.1 Background

In many developed nations, the banking sector has led the way in digital innovation and continues to enhance these services. Lebanon is experiencing a notable expansion in digital banking, evidenced by the active use of channels such as debit and credit cards, automated teller machines, home banking, and mobile banking. However, the transition to e-banking in Lebanon faces several obstacles. Service providers highlight issues such as limited professional expertise in delivering internet banking, insufficient customer awareness initiatives, and significant concerns around security. Despite these challenges, Lebanese banks are strategically leveraging advancements in e-banking to strengthen customer loyalty, investing substantially in modern digital strategies to preserve and expand their competitive edge.

1.2 Significance of the Study

The study's outcomes will support banks in optimizing their business practices while providing deeper insights into the issues surrounding digital banking services. The study will assist bankers and stakeholders in the financial sector in formulating strategies to advance digital banking services. In addition, it will serve as a valuable resource for future research, offering reference material that highlights both the challenges and opportunities associated with e-banking in Lebanon.

1.3 Importance of the Study

The importance of this study lies in the concept of digital banking and the factors that appear to be influencing the use of digital banking in Lebanon. The research helps banks identify the issues that must be considered to attract customers to adopt digital banking. The transition of banks to provide digital services in addition to traditional banking services has become an urgent need for banks to adapt to.

1.4 Objective of the Study.

This research aims to explore customers' views regarding digital banking services in Lebanon. More specifically, it seeks to assess the relative significance of various dimensions of service quality and examine how these factors

influence customers' perceptions and their level of engagement with electronic banking. Ultimately, the study intends to propose policy recommendations that address existing challenges from the standpoint of customers.

1.5 Literature Review

Previous research has attempted to examine the determinants of consumers' attitudes toward the digital banking sector, but there is still a lack of research in the context of the Lebanese banking sector. A study by (Asiyanbi & Ishola, March 2018) entitled "E-banking Services Impact and Customer Satisfaction in Selected Bank Branches in Nigeria. This study aimed to identify the impact of electronic banking services on customer satisfaction in Nigeria through a questionnaire distributed to 100 participants, which showed that 98% of customers use ATMs, 85% of them use online banking services, and 97% of customers benefit from electronic money transfers. It indicated that customers are satisfied and happy with electronic banking services because of their saving of time savings and ease of access, with some restrictions, which are the fear of bank fraud and the failure of electronic transactions. The study recommended the need to increase investment in the infrastructure of electronic banking services. Another study by Mei et al. (2016) entitled Effect of Electronic Banking on Customer Satisfaction in Kwara State, Nigeria. This study aimed to show that transaction speed is an important factor for e-banking customers. Transaction speed generally refers to the frequency of network connection interruptions, the time a customer spends responding to a page, and the bank's response rate to customers. Empirical evidence suggests that if this transaction time is shortened, individuals will opt for self-service, and time savings will become an important concept for those using Internet banking (Toor A., Hunain, Hussain, Ali, & Shahid, 2016).

Also, a study by Idowu and Fadiya (2015) entitled An Empirical Study of Automated Teller Machine Service Quality on Customer Satisfaction. The purpose of this paper was to measure customer satisfaction with e-banking services. This survey was conducted using a survey design that asked respondents about their e-banking services. The collected data were analyzed using multiple logistic regression analysis and Pearson's correlation. As a result, it was found that the necessary inputs to bank management are necessary to improve customer satisfaction through the improvement of e-banking service quality. Don't just focus on the quality aspect of e-banking. This aspect should complement and integrate with other aspects of the bank's service quality to satisfy its customers.

On the other hand, security aspects are less important than other aspects when measuring customer satisfaction related to e-banking services (Idowu & Fadiya, 2015).

The study of Al-Riyadi (2016), entitled "The Impact of Banking Services Quality Dimensions on Customer Satisfaction at Arab Bank" aimed to identify the impact of banking service quality on customer satisfaction. The results showed that customers are satisfied with the service provided by the bank, and safety was the best value in terms of customer satisfaction with it, because of the policies followed by the bank to make the customer feel safe. On the other hand, the results showed that the tangible variable is the least effective for achieving customer satisfaction among the other variables.

1.6 Problem Statement

This study seeks to evaluate and identify the factors influencing customer perceptions of digital banking engagement in Lebanon. The goal is to enhance service quality, uncover deficiencies, and propose solutions to address them. The research problem can be drawn up with the following questions:

Main Question:

Is there an impact of digital banking on customers' perception of online banking at Lebanese banks?

Sub-Problems:

- 1) Does tangibility (the physical aspects and appearance of digital banking services) influence customer perception of online banking?
- 2) Does reliability (the consistency and accuracy of digital banking services) affect customer perception?
- 3) Does responsiveness (the speed and efficiency of service delivery) impact customer perception?
- 4) Does assurance (the confidence and trust customers feel in using digital banking) shape customer perception?
- 5) Does empathy (the degree of personalized care and attention provided through digital banking) affect customer perception?

1.7 Hypothesis of the Study

The research aims at validating the following Hypotheses:

Null Hypothesis (H_0):

There is no statistically significant effect of the dimensions of digital-banking service quality (tangibility, reliability, responsiveness, assurance, and empathy) on customers' perception of the banks under study.

Alternative Hypothesis (H_1):

There is a statistically significant effect of the dimensions of digital-banking service quality (tangibility, reliability, responsiveness, assurance, and empathy) on customers' perception of the banks under study.

Sub-Hypotheses

Sub Hypothesis 1 – Reliability

H_0 : Reliability, as a dimension of digital-banking service quality, has no statistically significant effect on customers' perception of the banks under study.

H_1 : Reliability, as a dimension of digital-banking service quality, has a statistically significant effect on customers' perception of the banks under study.

Sub Hypothesis 2 – Tangibility

H_0 : Tangibility, as a dimension of digital-banking service quality, has no statistically significant effect on customers' perception of the banks under study.

H_1 : Tangibility, as a dimension of digital-banking service quality, has a statistically significant effect on customers' perception of the banks under study.

Sub Hypothesis 3 – Responsiveness

H_0 : Responsiveness, as a dimension of digital-banking service quality, has no statistically significant effect on customers' perception of the banks under study.

H_1 : Responsiveness, as a dimension of digital-banking service quality, has a statistically significant effect on customers' perception of the banks under study.

Sub Hypothesis 4 – Assurance

H_0 : Assurance, as a dimension of digital-banking service quality, has no statistically significant effect on customers' perception of the banks under study.

H_1 : Assurance, as a dimension of digital-banking service quality, has a statistically significant effect on customers' perception of the banks under study.

Sub Hypothesis 5 – Empathy

H_0 : Empathy, as a dimension of digital-banking service quality, has no statistically significant effect on customers' perception of the banks under study.

H_1 : Empathy, as a dimension of digital-banking service quality, has a statistically significant effect on customers' perception of the banks under study.

2. Methodology

This research will use quantitative methodology and multiple linear regression to analyze the data. Modeling the linear relationship between the explanatory independent factors and response-dependent variables is the aim of multiple linear regression. To complete the study, questionnaires were created using Google Forms software. After the completion of the data collection, the data will be exported as an Excel file and inputted into the SPSS statistical software version 26.0 to complete the data analysis.

2.1 Questionnaire

The questionnaire will consist of six different sections. The first part is about demography. The second section consists of five sections, five of which represent the independent variables (the five dimensions that are tangibility, reliability, responsiveness, assurance, and empathy) measured through the 5-point Likert scale, and one section representing the dependent variable (customer engagement), also measured through the 5-point Likert scale. Adding to that, the demographics section will be explored in order to obtain the descriptive statistics of the study.

2.2 Population

The study population consists of commercial banks operating in Lebanon. From this population, a random sample of 300 customers was selected across 40 different banks. Questionnaires were distributed to customers holding accounts

in both local and foreign currencies.

2.3 Sample Selection

A total of 500 questionnaires were distributed. Of these, 450 were returned, but 150 were deemed unusable due to damage or incompleteness. Consequently, 300 valid questionnaires were retained for analysis. The data collection process was carried out over one month.

2.4 Variables

The variables used were:

Dependent Variable (DV):

Customer perception and engagement

This represents how customers view, interact with, and commit to digital banking services in Lebanese banks.

Independent Variables (IVs):

Tangibility – The physical and visual aspects of digital banking services (e.g., ATM design, app interface).

Reliability – The consistency, accuracy, and dependability of digital banking services.

Responsiveness – The speed and efficiency with which banks address customer needs and issues.

Assurance – The level of trust, confidence, and security customers feel when using digital banking.

Empathy – The degree of personalized care, understanding, and support provided to customers.

3. Measurements

Regression analysis will be used to identify the relationships between explanatory variables and our main dependent variable. In this study, reliability analysis was used to establish the consistency and stability of the research instrument. Cronbach's alpha is a coefficient that indicates how well the variables in a set are positively correlated to one another. Also, linear correlation, ANOVA, coefficients, and multiple linear regression.

3.1 Descriptive Statistics

3.1.1 Data Analysis

Table 1. One-sample statistics of tangibility

	N	Mean	Std. Deviation	Std. Error Mean
Tangibility: The service is fast.	300	4.35	.737	.043
Tangibility: All transactions can be quickly completed	300	4.03	.867	.050
Tangibility: The service is easy to use.	300	4.21	.746	.043
Tangibility: The services support multiple languages	300	4.20	.764	.044
Tangibility: Learning how to use electronic banking is easy	300	4.19	.782	.045
Tangibility: The system is flexible in terms of interaction.	300	4.11	.785	.045

This table shows the mean and standard deviation for the 6 questions of tangibility.

The results show that Q3 has the highest mean (4.20), Q2 has the lowest mean (4.03), Q2 has the highest standard deviation (0.050), and Q1 and Q2 have the lowest standard deviation (0.043).

Table 2. One-sample statistics of reliability

	N	Mean	Std. Deviation	Std. Error Mean
Reliability: Reply within the specified time.	300	4.10	.909	.052
Reliability: Comfort when problems arise.	300	3.96	.874	.050
Reliability: Reliability in E-banking.	300	4.02	.871	.050
Reliability: The service was provided at the promised time .	300	4.15	.732	.042
Reliability: Accurate records.	300	4.20	.724	.042
Reliability: Transactions on the website are confidential.	300	4.18	.782	.045

This table shows the mean and standard deviation for the six questions of reliability. The results show that Q5 has the

highest mean (4.20), Q2 has the lowest mean (3.96), Q1 has the highest standard deviation (0.909), and Q5 has the lowest standard deviation (0.724).

Table 3. One-sample statistics of responsiveness

	N	Mean	Std. Deviation	Std. Error Mean
Responsiveness: The e-banking notifies the clients when the service will take place.	300	4.22	.825	.048
Responsiveness of e -banking service.	300	4.14	.745	.043
Responsiveness: ready to help customers anytime.	300	4.17	.774	.045
Responsiveness: Ready to respond to requests.	300	4.11	.773	.045
Responsiveness: Online services are available 24/7	300	4.13	.777	.045
Responsiveness: The website is easy to use.	298	4.15	.751	.044

The table shows the mean and standard deviation for the 6 questions of responsiveness.

The results show that Q1 has the highest mean (4.22) and Q4 has the lowest mean (4.11), as well as Q1 having the highest standard deviation (0.825) and Q2 having the lowest standard deviation (0.745).

Table 4. One-sample statistics of assurance

	N	Mean	Std. Deviation	Std. Error Mean
Assurance: The e-banking that instills confidence in customers.	300	4.31	.690	.040
Assurance: customers feel safe in their electronic transactions.	300	4.10	.731	.042
Assurance: E-banking is consistently polite.	300	4.17	.722	.042
Assurance has all the answers for clients that need it	300	4.04	.760	.044
Assurance: knowledgeable to answer customer questions	300	4.10	.769	.044
Assurance: services are secure and protected from fraud	300	4.07	.756	.044

This table shows the mean and standard deviation for the 6 questions of assurance. The results show that Q1 has the highest mean (4.31) and Q4 has the lowest mean (4.04), as well as Q4, Q5, and Q6 have the highest standard deviation (0.044), and Q1 has the lowest standard deviation (0.040).

Table 5. One-sample statistics of empathy

	N	Mean	Std. Deviation	Std. Error Mean
Empathy: gives clients personalized attention	300	4.18	.715	.041
Empathy: deals with clients with care.	300	4.10	.685	.040
Empathy: Focus on clients' well-being.	300	4.05	.854	.049
Empathy: Understands the client's needs	300	4.11	.742	.043
Empathy: Appropriate business hours.	300	4.19	.718	.041
Empathy: The bank's helpdesk and call center are empathetic.	300	4.10	.820	.047

This table shows the mean and standard deviation for the 6 questions on empathy. The results show that Q5 has the highest mean (4.19), Q3 has the lowest mean (4.05), Q3 has the highest standard deviation (0.854), and Q1 has the lowest standard deviation (0.715).

Table 6. One-sample statistics of the customer

	N	Mean	Std. Deviation	Std. Error Mean
I am able to perceive the transaction processing by the online banking service.	300	4.23	.751	.043
I believe I have decided to use digital banking services at my bank	300	4.11	.663	.038
My perception and engagement with digital banking services is high.	300	4.16	.695	.040
I can easily access the services provided online	300	4.06	.788	.046
The digital-banking service is easy to engage	300	4.09	.755	.044
The user-friendly characteristic of digital-banking helps in the adoption and perception of process.	300	4.15	.698	.040

This table shows the mean and standard deviation for the 6 questions of customer perception of online services. The results show that Q1 has the highest mean (4.23), Q4 has the lowest mean (4.09), Q4 has the highest standard deviation (0.788), and Q2 has the lowest standard deviation (0.633).

4. Discussing the Hypothesis

4.1 Reliability Analysis

Cronbach's alpha ranges from 0 to 1. The highest values indicate that the relationships between the items are stronger. A Cronbach's alpha of 0.7 or higher is usually considered acceptable.

Table 7. Reliability statistics

Reliability Statistics			
	Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
<i>Tangibility</i>	.886	.886	2
<i>Reliability</i>	.887	.889	2
<i>Responsiveness</i>	.895	.897	2
<i>Assurance</i>	.917	.917	2
<i>Empathy</i>	.956	.956	2

The table indicates that most variables demonstrate relatively high Cronbach's alpha values, ranging from 0.886 to 0.956, which reflects medium to high reliability across the scale. Among these, the tangibility dimension records a Cronbach's alpha of 0.886, suggesting comparatively lower internal consistency relative to the other variables, though still within an acceptable range. In contrast, the empathy dimension achieves the highest value at 0.956, signifying a very strong level of internal consistency.

4.2 Multicollinearity

Table 8. Multicollinearity

Model	Unstandardized Coefficients		Standardized Coefficients	T	Sig.	Correlations			Collinearity Statistics	
	B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
(Constant)	.120	.093		1.289	.198					
Tangibility	.163	.035	.160	4.624	.000	.795	.260	.093	.339	2.949
Reliability	.000	.038	.000	.010	.992	.800	.001	.000	.239	4.185
1 Responsive	.140	.036	.152	3.855	.000	.813	.219	.078	.264	3.792
Assurance	.154	.042	.154	3.664	.000	.846	.209	.074	.232	4.319
Empathy	.513	.043	.544	11.943	.000	.917	.572	.241	.197	5.085

Variance Inflation Factor (VIF) values were calculated for the five independent variables representing dimensions of digital-banking service quality. VIF begins at 1, with higher values indicating stronger correlations among predictors:

VIF = 1 → No correlation with other variables.

VIF between 1 and 5 → Moderate correlation, generally acceptable without corrective action.

VIF above 5 → High multicollinearity, suggesting unstable coefficient estimates and questionable p-values.

Results

Tangibility: VIF = 2.949 → Moderate correlation, acceptable.

Reliability: VIF = 4.185 → Moderate correlation, acceptable.

Responsiveness: VIF = 3.792 → Moderate correlation, acceptable.

Assurance: VIF = 3.792 → Moderate correlation, acceptable.

Empathy: VIF = 5.085 → Exceeds the threshold, indicating problematic multicollinearity and less reliable coefficient estimation.

The analysis shows that most variables fall within the acceptable range (1–5), suggesting moderate but manageable correlations. However, empathy exceeds the threshold, implying that its effect may overlap substantially with other dimensions, leading to instability in regression estimates. This issue may require corrective measures such as variable transformation, removal, or the use of advanced techniques (e.g., ridge regression or principal component analysis) to address multicollinearity.

4.3 Correlation

Table 9. Correlation coefficient

		Tangibility	Reliability	Responsiveness	Assurance	Empathy	Customer
Tangibility	Pearson Correlation	1	.755**	.747**	.689**	.763**	.795**
	Sig. (2-tailed)		.000	.000	.000	.000	.000
	N	300	300	300	300	300	300
Reliability	Pearson Correlation	.755**	1	.819**	.784**	.797**	.800**
	Sig. (2-tailed)	.000		.000	.000	.000	.000
	N	300	300	300	300	300	300
Responsiveness	Pearson Correlation	.747**	.819**	1	.762**	.778**	.813**
	Sig. (2-tailed)	.000	.000		.000	.000	.000
	N	300	300	300	300	300	300
Assurance	Pearson Correlation	.689**	.784**	.762**	1	.857**	.846**
	Sig. (2-tailed)	.000	.000	.000		.000	.000
	N	300	300	300	300	300	300
Empathy	Pearson Correlation	.763**	.797**	.778**	.857**	1	.917**
	Sig. (2-tailed)	.000	.000	.000	.000		.000
	N	300	300	300	300	300	300
Customer	Pearson Correlation	.795**	.800**	.813**	.846**	.917**	1
	Sig. (2-tailed)	.000	.000	.000	.000	.000	
	N	300	300	300	300	300	300

* Correlation is significant at the 0.05 level (2-tailed).

** Correlation is significant at the 0.01 level (2-tailed).

The results of the correlation analysis demonstrate strong positive and statistically significant relationships between each dimension of digital-banking service quality and customer perception:

- Tangibility → Customer Perception: $r=0.795$, $p<0.05$
- Reliability → Customer Perception: $r=0.800$, $p<0.05$
- Responsiveness → Customer Perception: $r=0.813$, $p<0.05$
- Assurance → Customer Perception: $r=0.849$, $p<0.05$
- Empathy → Customer Perception: $r=0.917$, $p<0.05$

All five dimensions exhibit strong positive correlations with customer perception, confirming that improvements in service quality are closely associated with more favorable customer perceptions of digital banking.

Among the variables, empathy shows the strongest correlation ($r= 0.917$), suggesting that personalized care and customer-focused support play a particularly critical role in shaping perceptions.

Assurance also demonstrates a strong correlation ($r= 0.849$), highlighting the importance of trust and confidence in digital banking services.

While tangibility has the lowest correlation ($r = 0.795$), it remains statistically significant and indicates that the physical and interface aspects of digital banking still contribute meaningfully to customer perception.

4.4 Regression Analysis

4.4.1 Model Summary

The regression model was constructed using the five dimensions of digital-banking service quality (tangibility, reliability, responsiveness, assurance, and empathy) as independent variables, with customer perception as the dependent variable.

R^2 value: Indicates the proportion of variance in customer perception explained by the five dimensions. A high R^2 suggests strong explanatory power of the model.

Adjusted R^2 : Provides a more accurate measure by adjusting for the number of predictors, ensuring the model is not over-fitted.

4.4.2 Coefficients and Significance

The regression coefficients (β) and p-values for each dimension reveal the relative strength and statistical significance of their impact on customer perception:

Table 10. Coefficients and significance

Dimension	β Coefficient	Significance (p-value)	Interpretation
Tangibility	Positive, significant	$p < 0.05$	Tangibility contributes meaningfully to customer perception, though less strongly than other dimensions.
Reliability	Positive, significant	$p < 0.05$	Reliability strongly influences customer perception, confirming that consistent service builds trust.
Responsiveness	Positive, significant	$p < 0.05$	Responsiveness has a notable effect, highlighting the importance of timely service delivery.
Assurance	Positive, significant	$p < 0.05$	Assurance shows a very strong effect, emphasizing the role of confidence and security in digital banking.
Empathy	Highest positive, significant	$p < 0.05$	Empathy demonstrates the strongest effect, indicating that personalized care and customer-focused support are critical drivers of perception.

Source: prepared by researcher.

The regression analysis confirms that all five dimensions significantly predict customer perception of digital banking services in Lebanese banks.

Empathy emerges as the most influential predictor, followed by assurance, suggesting that emotional connection and trust are central to shaping customer attitudes.

Tangibility, while significant, has the weakest effect, implying that physical and interface aspects are less critical compared to relational and trust-based factors.

The overall model demonstrates strong explanatory power, validating the research hypothesis that service quality dimensions significantly affect customer perception.

Multiple regression analysis was used in this study to determine the predictive values of the constructs under consideration.

Table 11. Model summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.938 ^a	.880	.878	.21120

a. Predictors: (Constant), Empathy, Tangibility, Responsiveness, Reliability, Assurance

b. Dependent Variable: Customer perception.

Table 12. ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	96.180	5	19.236	431.247	.000 ^b
	Residual	13.114	294	.045		
	Total	109.294	299			

a dependent Variable: Customer perception.

b Predictors: (Constant), Empathy, Tangibility, Responsiveness, Reliability, Assurance.

The model summary indicates that the five dimensions of digital-banking service quality—empathy, reliability, responsiveness, tangibility, and assurance—explain a substantial proportion of the variance in customer perception. The R^2 value of 0.88 demonstrates that 88% of the variance in customer perception can be predicted from these independent variables, reflecting a very strong explanatory power of the model.

The ANOVA test confirms the overall significance of the regression model. With $F = 431.247$ and $p < 0.05$, the model is statistically significant. This result validates that the five service quality dimensions collectively have a meaningful impact on customer perception.

The high R^2 value suggests that the model is highly effective in explaining customer perception of digital banking services in Lebanese banks.

The significant ANOVA result indicates that the regression equation is a good fit for the data.

Each of the five dimensions—empathy, reliability, responsiveness, tangibility, and assurance—contributes significantly to predicting customer perception, supporting the study's main hypothesis.

Table 13. Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Correlations			Collinearity Statistics	
	B	Std. Error	Beta			Lower Bound	Upper Bound	Zero-order	Partial	Part	Tolerance	VIF
(Constant)	.120	.093		1.289	.198	-.063	.304					
Tangibility	.163	.035	.160	4.624	.000	.094	.233	.795	.260	.093	.339	2.949
Reliability	.000	.038	.000	.010	.992	-.075	.076	.800	.001	.000	.239	4.185
Responsive	.140	.036	.152	3.855	.000	.068	.211	.813	.219	.078	.264	3.792
Assurance	.154	.042	.154	3.664	.000	.071	.236	.846	.209	.074	.232	4.319
Empathy	.513	.043	.544	11.943	.000	.428	.597	.917	.572	.241	.197	5.085

a dependent Variable: Customer perception.

b Predictors: (Constant), Empathy, Tangibility, Responsiveness, Reliability, Assurance.

4.4.3 Regression Coefficients Analysis

The regression coefficients provide insight into the relative contribution of each service quality dimension to customer perception.

Table 14. Regression coefficient analysis

Variable	B (Unstandardized)	Beta (Standardized)	t-value	Sig. (p)	Interpretation
Tangibility	0.163	0.160	4.624	0.000	Significant positive effect; tangibility contributes moderately to customer perception.
Reliability	0.000	0.000	0.010	0.992	Not significant; reliability does not predict customer perception in this model.
Responsiveness	0.140	0.152	3.855	0.000	Significant positive effect; responsiveness plays a meaningful role in shaping perception.
Assurance	0.154	0.154	3.664	0.000	Significant positive effect; assurance strongly influences customer trust and perception.
Empathy	0.513	0.544	11.943	0.000	Most significant predictor; empathy has the strongest impact on customer perception.

a dependent Variable: Customer perception

b Predictors: (Constant), Empathy, Tangibility, Responsiveness, Reliability, Assurance Empathy is the most influential predictor ($\beta = 0.544$, $p < 0.001$), highlighting the critical importance of personalized care and customer-focused support in digital banking.

Assurance and tangibility also show significant positive effects, reinforcing the role of trust, security, and the physical/interface aspects of digital banking.

Responsiveness contributes significantly, emphasizing the importance of timely service delivery.

Reliability, however, is not statistically significant ($p = 0.992$), suggesting that consistency and accuracy of services may not be perceived as a differentiating factor by Lebanese customers in this context.

The overall model remains highly explanatory ($R^2 = 0.88$), but the results indicate that empathy and assurance are the strongest drivers of customer perception, while reliability does not add predictive value.

5. Hypothesis Result

This section presents the results of the multiple linear regressions to confirm or reject the hypotheses related to the

five dimensions of digital-banking quality services on customer perception in Lebanese banks. In our case, there will be one model of multiple regression. For each, the three main tables of multiple regression will be drawn and analyzed to get a deep understanding and better results. Under each table, there will be a brief explanation of the specific table, and then, after drawing the three tables, there will be a deep explanation of the results.

Model: The impact of five dimensions on customer perception

$$Y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4 + \beta_5X_5 + \varepsilon t$$

Where:

Y = Customer perception. β_0 = Constant.

$\beta_1, \beta_2, \beta_3$ = Regression coefficients. X1 = Tangible.

X2 = Reliability.

X3 = Responsiveness.

X4=Assurance.

X5=Empathy.

εt = Error term.

Table 15. Model summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.938 ^a	.880	.878	.21120

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b. Dependent Variable: Customer perception

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	Residual	13.114	294	.045		
	Total	109.294	299			

a. Dependent Variable: Customer perception

b. Predictors: (Constant), Empathy, Tangibility, Responsiveness, Reliability, Assurance

Table 17. Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
		B	Std. Error				Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	.120	.093		1.289	.198					
	Tangibility	.163	.035	.160	4.624	.000	.795	.260	.093	.339	2.949
	Reliability	.000	.038	.000	.010	.992	.800	.001	.000	.239	4.185
	Responsiveness	.140	.036	.152	3.855	.000	.813	.219	.078	.264	3.792
	Assurance	.154	.042	.154	3.664	.000	.846	.209	.074	.232	4.319
	Empathy	.513	.043	.544	11.943	.000	.917	.572	.241	.197	5.085

a dependent Variable: Customer perception.

b Predictors: (Constant), Empathy, Tangibility, Responsiveness, Reliability, Assurance.

The independent variables collectively provide a strong explanatory power, indicating that service quality dimensions are highly predictive of customer perception.

$R^2 = 0.88 \rightarrow$ The model explains 88% of the variance in customer perception.

Adjusted R^2 (slightly lower) confirms the robustness of the model without overfitting.

$F = 431.247, p < 0.05 \rightarrow$ The regression model is statistically significant.

The overall model is valid, meaning that the five dimensions of service quality jointly have a significant effect on

customer perception.

To analyze the hypothesis, we used multiple linear regression analysis at 95% confidence intervals. The analysis showed $R^2 = 0.880$ and adjusted $R^2 = 0.878$. The analysis shows that empathy had a positive effect on customer perception ($\beta = 0.544$, $t = 11.943$, $p < 0.001$). Hence, the hypothesis was accepted.

The analysis shows that assurance had a positive effect on customer perception ($\beta = 0.154$, $t = 4.624$, $p < 0.001$). Hence, the hypothesis was accepted. The analysis shows that tangibility had a positive effect on customer perception ($\beta = 0.160$, $t = 4.624$, $p < 0.001$). Hence, the hypothesis was accepted.

The analysis shows that responsiveness had a positive effect on customer perception $\beta = 0.152$, $t = 3.855$, $p < 0.001$). Hence, the hypothesis was accepted. The analysis shows Reliability did not affect customer perception ($\beta = 0.000$, $t = 0.010$, $p > 0.001$). Hence, the hypothesis was rejected.

The model equation:

$$\text{Customer perception} = 0.120 + 0.163 (\text{tangibility}) + 0.140 (\text{responsiveness}) + 0.154 (\text{assurance}) + 0.544 (\text{empathy})$$

According to the findings, the accepted and rejected hypotheses are as follows:

Accepted Hypotheses

1. H₁: Tangibility has a statistically significant effect on customer perception of digital-banking services.
2. H₁: Responsiveness has a statistically significant effect on customer perception of digital-banking services.
3. H₁: Assurance has a statistically significant effect on customer perception of digital-banking services.
4. H₁: Empathy has a statistically significant effect on customer perception of digital-banking services.

Rejected Hypothesis

5. H₀: Reliability has no statistically significant effect on customer perception of digital-banking services.

Table 18. Hypothesis results

Variable	B (Unstandardized)	Beta (Standardized)	t-value	Sig. (p)	VIF	Interpretation
Constant	0.120	—	1.289	0.198	—	Not significant
Tangibility	0.163	0.160	4.624	0.000	2.949	Significant positive effect
Reliability	0.000	0.000	0.010	0.992	4.185	Not significant
Responsiveness	0.140	0.152	3.855	0.000	3.792	Significant positive effect
Assurance	0.154	0.154	3.664	0.000	4.319	Significant positive effect
Empathy	0.513	0.544	11.943	0.000	5.085	Strongest significant effect

Source: prepared by researcher.

6. Conclusion

This study examined the impact of digital-banking service quality on customer perception in Lebanese banks, focusing on five dimensions: tangibility, reliability, responsiveness, assurance, and empathy.

Reliability and validity tests confirmed that all variables were statistically reliable, with Cronbach's alpha values exceeding 0.7. Multiple regression analysis revealed that tangibility, responsiveness, assurance, and empathy significantly influence customer perception, while reliability showed no significant effect.

Among the predictors, empathy emerged as the strongest driver, followed by assurance, tangibility, and responsiveness. The findings highlight that customers value ease of use, time-saving features, trust, and personalized support in digital banking services. Interestingly, reliability was not a significant factor, suggesting that customers may already expect reliability as a baseline requirement, or that concerns about privacy, security, and complex authentication procedures reduce its perceived importance.

Overall, the study confirms a strong positive correlation between digital-banking service quality and customer perception, with an explanatory power of 88% ($R^2 = 0.88$).

The results of the multiple regression analysis confirm that four dimensions of digital-banking service quality—tangibility, responsiveness, assurance, and empathy—have a statistically significant impact on customer perception in Lebanese banks. These variables emerged as the most influential predictors, highlighting the importance of user-friendly interfaces, timely service, trust and security, and personalized support in shaping

customer attitudes toward digital banking. The findings also revealed that reliability does not significantly affect customer perception. This suggests that customers may already consider reliability a baseline expectation rather than a differentiating factor. In practice, customers continue to use electronic services even when they perceive them as less reliable, possibly due to convenience and time-saving benefits.

7. Recommendations

Based on the findings, the following recommendations are proposed for Lebanese banks:

Enhance Empathy and Customer-Centric Support.

Prioritize personalized communication, customer care, and tailored digital solutions.

Invest in AI-driven chatbots and customer service platforms that provide empathetic, human-like support.

Strengthen Assurance and Security; Improve transparency in security protocols to build customer trust. Simplify authentication processes (e.g., biometric login, two-factor authentication) to reduce frustration while maintaining safety.

Improve Tangibility and User Experience; Design user-friendly mobile apps and online platforms with intuitive interfaces.

Ensure consistency across digital touchpoints (ATMs, apps, websites) to reinforce brand trust.

Boost Responsiveness; Provide faster resolution of customer issues through real-time support channels.

Implement proactive notifications (e.g., transaction alerts, fraud detection) to reassure customers.

Address Reliability Concerns; Although reliability was not statistically significant, banks should not ignore it. Conduct awareness campaigns to educate customers about system reliability and data protection. Simplify login processes and reduce technical barriers that discourage usage.

Lebanese banks should focus on empathy, assurance, tangibility, and responsiveness as the key drivers of customer perception in digital banking. By addressing customer anxieties around reliability and privacy, banks can further strengthen engagement and trust, ensuring wider adoption of digital services.

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